

GSM Based Gas Leakage Detection and Alert System

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INTRODUCTION

- *An Arduino Uno gas leakage detection system is a low-cost, automated safety device using sensors (like MQ-2, MQ-5) to monitor hazardous gases (LPG, methane, smoke) and alert users through buzzers, LEDs, and optionally, SMS alerts (GSM module) or IoT, preventing accidents by detecting leaks and triggering alarms or even shutting off gas valves automatically. It's a practical project for enhancing safety in homes and industries, providing timely warnings and reducing reliance on manual checks.*

It is useful for early detection of gas leaks, safety enhancement and environment monitoring

OBJECTIVES

- **Enhance Safety & Prevent Accidents:** *The primary goal is to stop catastrophic events like fires, explosions, and asphyxiation by detecting leaks before they escalate.*
- **Early Warning System:** *To identify even small traces of gas, providing immediate alerts (audible alarms, LEDs, SMS) for prompt action.*
- **Minimize Risks:** *Reduce the likelihood of fatalities, injuries, and extensive property damage in domestic and industrial settings.*
- **Automated Response:** *Trigger automatic actions, such as shutting off the gas supply, for instant hazard mitigation.*
- **Monitor Toxic & Flammable Gases:** *Detect harmful substances like CO, methane, and H₂S, as well as combustible gases (LPG).*
- **Provide Real-Time Data:** *Offer continuous monitoring and remote notifications, often via IoT, for better control and oversight.*
- **Protect Vulnerable Environments:** *Safeguard homes, workplaces, oil rigs, and confined spaces where gas leaks pose significant threats.*

REQUIREMENTS

COMPONENTS	DESCRIPTION
ARDIUNO UNO(R3)	<i>Main controller to read sensor input and output</i>
MQ- SENSORS <i>Eg:MQ-2,MQ-6,etc</i>	<i>Detects harmful gas like LPG,METHANE and sends a analog signals to Arduino</i>
16*2 LCD WITH I2C MODULE	<i>Displays real time gas levels and system status</i>
GREEN LED	<i>Indicates safe gas levels</i>
RED LED	<i>Indicates different gas levels</i>
BUZZER/ALARM	<i>Sounds when the gas concentration reaches/exceeds the threshold</i>
JUMPER WIRES	<i>For connecting components to Arduino uno</i>
BREAD BOARD	<i>Used for prototyping the circuits</i>
POWER SUPPLY	<i>Powers Arduino and connected devices</i>
GSM MODULE SIM800A	SIM800A is a 2G GSM/GPRS module used to make calls, send SMS

SENSORS AND ACTUATORS

SENSORS ;

MQ-2 GAS SENSOR- *detects the precence and concentration of combustibile and harmful gases and sens an analog voltage signal to Arduino*



ACTUATORS

- **LED INDICATORS:** green led indicates safe gas level and red led indicates dangerous gas level



- **BUZZER OR ALARM:** sounds when gas concentration exceeds the threshold level



- **16*2 DISPLAY WITH I2C CD MODULE:** *Displays the real time gas levels and system status*



- **SERVO MOTOR(SG-90):** *used to control regulator(OFF/ON) in LPG gas cylinder*



GSM MODULE (SIM 800A)

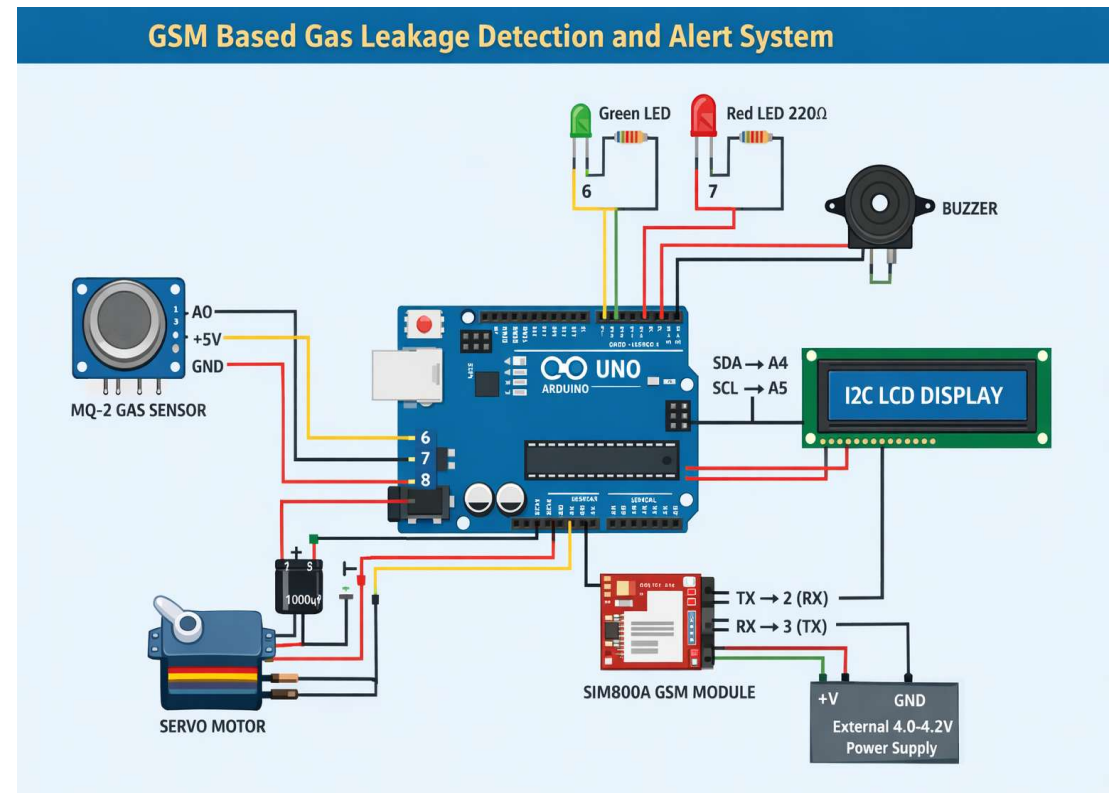
*The **SIM 800A** is a compact, cost-effective 2G GSM/GPRS module designed for IoT applications, enabling voice calls, SMS, and data connectivity via AT commands*



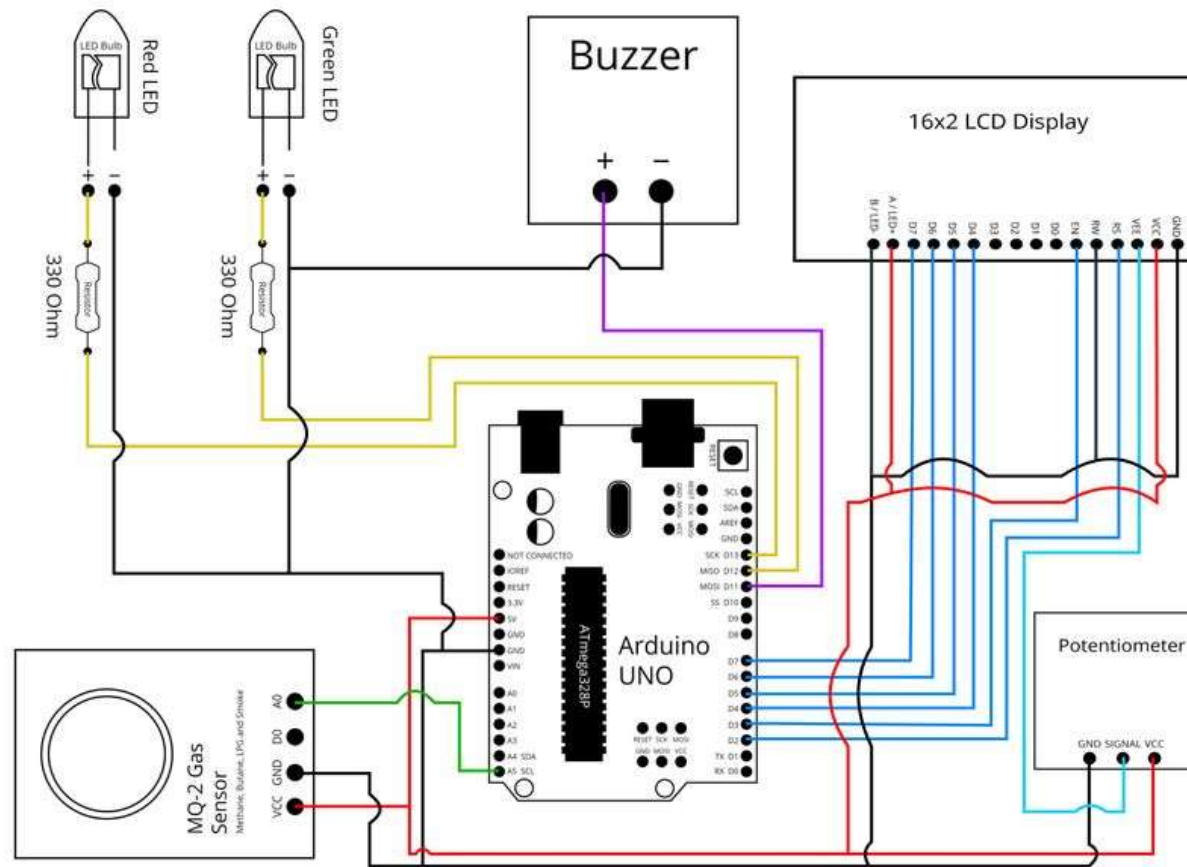
HARDWARE SETUP

The system uses:

1. *Ardiuno uno as a main controller*
2. *MQ gas sensor for detecting harmful gases*
3. *16*2 LCD display gas levels and system status*
4. *Green and red led for visual indications*
5. *Buzzer for audible alerts*
6. *GSM used to send SMS and CALL alert to phone*



CIRCUIT DIAGRAM



REAL TIME PROJECT SETUP



CODE IMPLEMENTATION

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <Servo.h>
#include <SoftwareSerial.h>

LiquidCrystal_I2C lcd(0x27, 16, 2);
Servo regulatorServo;
SoftwareSerial gsm(2, 3); // RX, TX

// Pins
int gasSensor = A0;
int buzzer = 8;
int redLed = 7;
int greenLed = 6;
int servoPin = 9;

// Settings
int thresholdPercent = 40;

bool alertActive = false;
bool callActive = false;
bool smsSent = false;

unsigned long gasStartTime = 0;

void setup() {
  pinMode(buzzer, OUTPUT);
  pinMode(redLed, OUTPUT);
  pinMode(greenLed, OUTPUT);

  regulatorServo.attach(servoPin);
  regulatorServo.write(0); // Regulator OPEN

  lcd.init();
  lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print("Gas Leakage");
  lcd.setCursor(0, 1);
  lcd.print("Detector Ready");
  delay(3000);
  lcd.clear();

  gsm.begin(9600);
  delay(3000);
```

```
  gsm.println("AT");
  delay(1000);
  gsm.println("AT+CMGF=1");
  delay(1000);
}

void loop() {
  int gasValue = analogRead(gasSensor);
  int gasPercent = map(gasValue, 0, 1023, 0, 100);

  lcd.setCursor(0, 0);
  lcd.print("Gas: ");
  lcd.print(gasPercent);
  lcd.print("% ");

  // GAS CONFIRMATION (2 sec)
  if (gasPercent > thresholdPercent) {
    if (gasStartTime == 0)
      gasStartTime = millis();

    if (millis() - gasStartTime > 2000) {
      gasAlert(gasPercent);
    }
  } else {
    gasStartTime = 0;
    gasSafe();
  }

  receiveSMS();
  delay(500);
}

void gasAlert(int gasPercent) {
  digitalWrite(buzzer, HIGH);
  digitalWrite(redLed, HIGH);
  digitalWrite(greenLed, LOW);
  regulatorServo.write(90); // Regulator CLOSE

  lcd.setCursor(0, 1);
  lcd.print("!! GAS ALERT !!");

  if (!smsSent) {
    sendSMS(gasPercent);
    smsSent = true;
  }
}
```

```
void gasSafe() {
  digitalWrite(buzzer, LOW);
  digitalWrite(redLed, LOW);
  digitalWrite(greenLed, HIGH);
  regulatorServo.write(0); // Regulator OPEN

  lcd.setCursor(0, 1);
  lcd.print("Status: SAFE ");

  if (callActive) {
    gsm.println("ATH"); // End call
    callActive = false;
  }

  smsSent = false;
  alertActive = false;
}

void sendSMS(int percent) {
  gsm.println("AT+CMGS=\"+91XXXXXXXXXX\"");
  delay(1000);
  gsm.print("Gas Leakage Detector Alert! Gas Level: ");
  gsm.print(percent);
  gsm.println("%");
  gsm.write(26);
  delay(3000);
}

void makeCall() {
  gsm.println("ATD+91XXXXXXXXXX");
}

void receiveSMS() {
  if (gsm.available()) {
    String msg = gsm.readString();
    msg.toUpperCase();

    if (msg.indexOf("OFF") != -1) {
      regulatorServo.write(90);
      lcd.clear();
      lcd.print("Regulator OFF");
    }

    if (msg.indexOf("ON") != -1) {
      regulatorServo.write(0);
      lcd.clear();
      lcd.print("Regulator ON");
    }
  }
}
```


METHODOLOGY

- **Gas Sensing**

*The MQ-2 gas sensor continuously senses the presence of combustible gases and smoke in the surrounding air. The sensor produces an **analog voltage output** proportional to the gas concentration.*

- **Gas Level Calculation**

*The sensed analog value is mapped into a **percentage gas level (0–100%)** using the Arduino program for easy understanding and display.*

- **Display of Gas Level**

*The calculated gas percentage and system status are displayed on a **16×2 I2C LCD**, showing real-time gas concentration and safety condition.*

- **Safe Condition**

When the gas level is below the threshold:

- The **green LED** remains ON
- The **buzzer** remains OFF
- The **servo motor** stays in normal position
- The LCD displays **SAFE status**

- **Continuous Monitoring**

The system continuously monitors gas concentration in a loop and updates outputs in real time to ensure safety.

FUTURE WORK

- *Automatic exhaust fan control can be added to improve ventilation during gas leakage.*
- *Multiple gas sensors can be used to detect different types of gases more accurately.*
- *Gas concentration data can be logged for monitoring and analysis over time.*
- *Mobile app or IoT dashboard can be implemented for remote monitoring and alerts.*
- *Replacing Arduino Uno and GSM with ESP32 enables Wi-Fi/Bluetooth, cloud monitoring, push notifications, and a smarter, scalable system*

RESULT

- *The system successfully detects gas leakage using the MQ-2 gas sensor and continuously monitors gas concentration.*
- *On gas detection, the buzzer and red LED activate immediately to warn nearby people.*
- *The GSM module sends alert notifications (SMS/CALL) to the user, enabling remote awareness from anywhere.*
- *The LCD displays real-time gas status, making the system easy to monitor locally.*
- *The project provides a reliable, low-cost safety solution for homes, industries, and LPG-based environments.*

CONCLUSION

- *The GSM Based Gas Leakage Detection and Alert System was successfully designed and implemented to provide early detection of gas leakage and ensure safety. The system continuously monitors gas levels using the MQ-2 sensor and responds instantly with visual (LED), audible (buzzer), and remote (SMS/call) alerts when a leak is detected. This helps in preventing accidents, fire hazards, and loss of life or property. The project proves to be cost-effective, reliable, and suitable for homes, industries, and commercial kitchens. Overall, the system demonstrates how embedded systems and communication technology can be effectively combined to enhance safety and automation.*